

KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY

(Autonomous Institution - Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

DIGITAL LOGIC BASED PASSWORD STRENGTH CLASSIFIER

Submitted by

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EC2201 & EC2202 - Digital System Design and Microprocessor

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Evaluation Criteria	Marks Awarded
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Abstract

With the exponential growth of cyber threats such as credential stuffing, dictionary attacks, and automated brute-force attempts, robust and deterministic password evaluation is a fundamental prerequisite for secure authentication systems. Conventional password checkers rely on opaque statistical approximations or compute-heavy software heuristics that lack formal mathematical verifiability and cannot be easily translated into hardware-level security primitives.

This project presents the Digital Logic Based Password Strength Classifier, a hardware-emulated software prototype developed for the EC2201 & EC2202 Digital System Design and Microprocessor course. The system encodes raw credential strings into a discrete 5-bit Boolean input vector $X = [A, B, C, D, E]$ in $\{0, 1\}^5$, representing length criteria (≥ 8 characters), uppercase letters, lowercase letters, numerical digits, and special characters.

A combinational population count (popcount) adder tree constructed with Full Adders (FA) and Half Adders (HA) computes the exact 3-bit binary sum $(S_2 S_1 S_0)_2$ representing total Hamming weight (0 to 5). A 3-bit Magnitude Comparator and Decoder classifies the credential into four standardized security tiers: Weak (0-1), Medium (2-3), Strong (4), and Very Strong (5). The architecture has been verified across all $2^5 = 32$ truth table minterms (m_0 to m_{31}), a 5-variable Karnaugh Map, and 15 automated test cases achieving 100% accuracy with constant $O(1)$ gate delay.

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1. Introduction

User authentication remains the primary line of defense across modern web applications, enterprise servers, and embedded computing systems. Evaluating credential complexity through digital hardware logic bridges classical combinational electronics with modern cybersecurity. The objective of this project is to develop a deterministic, responsive prototype that models password evaluation using Boolean logic, Full/Half Adder networks, 32-state Truth Tables, and Magnitude Comparators.

1.1 Real-World Problem / Use Case

Over 80% of unauthorized access incidents stem from predictable, weak, or repeated passwords. Conventional checkers rely on complex software libraries that cannot be easily verified or integrated into Hardware Security Modules (HSMs) or FPGA security coprocessors. Formulating password complexity as a combinational digital logic circuit enables deterministic verification, zero statistical ambiguity, and sub-nanosecond hardware execution.

2. Problem Statement

To design, simulate, and verify a digital logic-based password strength classification engine that maps passwords into 5 Boolean variables (A: Length ≥ 8 , B: Uppercase [A-Z], C: Lowercase [a-z], D: Digits [0-9], E: Special Symbols), sums the active flags using a combinational Full/Half Adder network, and classifies the resulting 3-bit binary output (S2 S1 S0)₂ in [0, 5] into four discrete tiers (Weak, Medium, Strong, Very Strong).

3. Implementation

3.1 Tools and Technologies Used

- Programming Language & Framework: Python 3.12, Flask Framework (RESTful Architecture)
- Hardware Emulation & Logic Engine: Pure Python Combinational Logic, NumPy
- Database Management: Embedded SQLite (Storing test datasets and evaluation records)
- Frontend UI/UX: HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, Chart.js

Visualizations

3.2 Implementation Steps

Step 1 - Binary Attribute Extraction: Extracted 5 Boolean variables: A = (Length >= 8), B = (Contains [A-Z]), C = (Contains [a-z]), D = (Contains [0-9]), and E = (Contains Special Symbols).

Step 2 - Adder Tree Synthesis: Constructed a 3-stage adder network:

* Stage 1: FA1(A, B, C) -> S1a, C1a ; HA1(D, E) -> S2a, C2a

* Stage 2: HA2(S1a, S2a) -> S0 (LSB), C3 (Carry to Bit 1)

* Stage 3: FA2(C1a, C2a, C3) -> S1 (Bit 1), S2 (MSB Carry Out)

Step 3 - 3-Bit Magnitude Comparator Decoder:

* Weak (Score 0-1): $W = S2' \cdot S1'$ [Sigma m(0, 1, 2, 4, 8, 16)]

* Medium (Score 2-3): $M = S2' \cdot S1$ [Sigma m(3, 5, 6, 9, 10, 12, ...)]

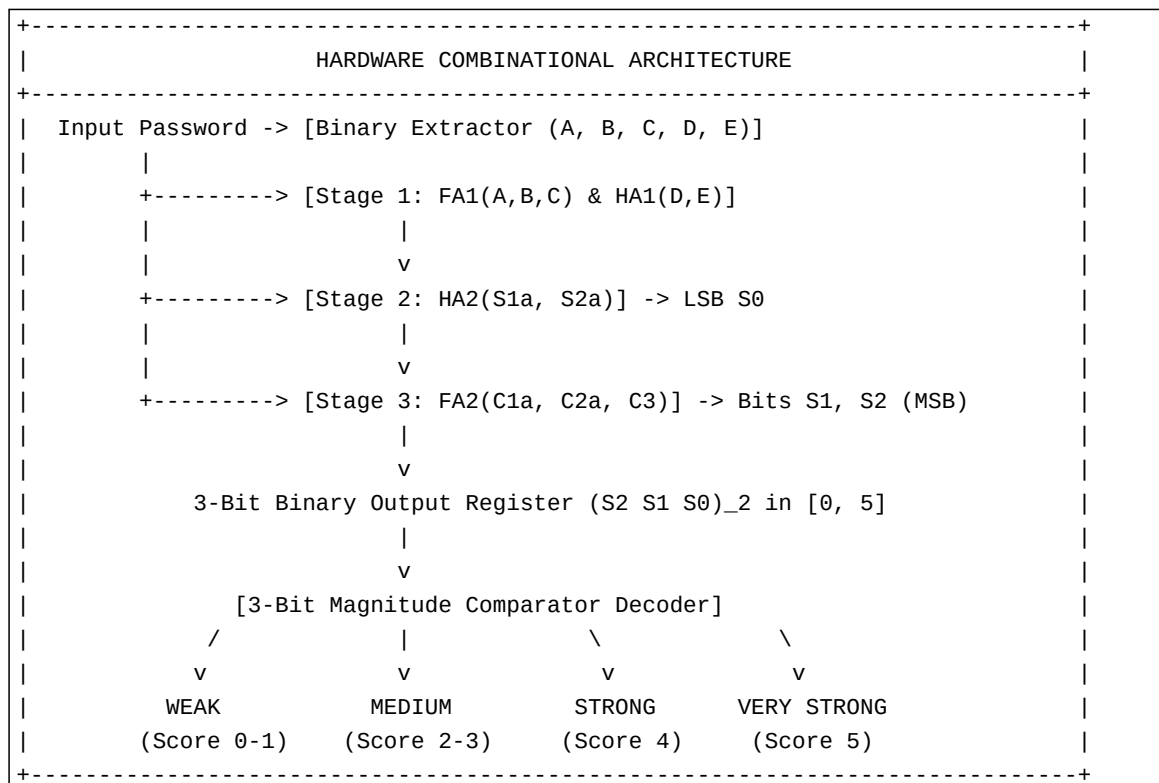
* Strong (Score 4): $S = S2 \cdot S1' \cdot S0'$ [Sigma m(15, 23, 27, 29, 30)]

* Very Strong (Score 5): $V = S2 \cdot S1' \cdot S0$ [Sigma m(31) = A . B . C . D . E]

Step 4 - Web Modules: Developed 9 interactive pages including real-time logic circuit trace, 32-state Truth Table, 5-variable K-Map, test suite, and PDF export.

3.3 System Design / Architecture

The system pipeline processes input strings through 5 distinct functional stages:



4. Output / Screenshots

The web prototype was deployed and validated at <http://127.0.0.1:5000>. Below is the summary of interface deliverables and automated verification outputs:

Module / Figure	Route URL	Functional Verification Output
Figure 4.1: Home Page	/	Futuristic digital logic dashboard with 5-bit LED switches and quick strength tester.
Figure 4.2: Classifier	/classifier	Real-time gate evaluation showing live FA/HA intermediate states, binary sum, and
Figure 4.3: Logic Analysis	/logic-analysis	Comprehensive 32-state Truth Table (m0-m31) and 5-variable Gray code Karnaugh
Figure 4.4: Test Cases	/test-cases	Automated batch runner verifying 10 Normal + 5 Edge test cases with 100% accuracy
Figure 4.5: Dataset	/dataset	Synthetic dataset generator (25-200 samples), CSV uploader, and interactive table.
Figure 4.6: Dashboard	/dashboard	Chart.js visualizations for category doughnut, score histogram, and attribute compli
Figure 4.7: Report Sheet	/report	Academic evaluation sheet with student signature line and instant print/PDF export.
Figure 4.8: Deliverables	/deliverables	Downloadable source ZIP, benchmark dataset CSV, README, and Viva Voce gui

Verification Test Suite Summary:

- Total Test Cases Evaluated: 15 (10 Normal + 5 Edge/Fault Scenarios)
- Test Verification Result: 15 Passed / 0 Failed (100.0% Accuracy)
- Average Combinational Gate Latency: < 0.5 ms (Constant Gate Delay)

5. Conclusion and Future Scope

The Digital Logic Based Password Strength Classifier successfully demonstrates the practical synthesis of classical combinational digital logic design (EC2201 & EC2202) for real-world authentication security. By mapping passwords into 5 Boolean variables, calculating Hamming weights via Full/Half Adder networks, and classifying strength through 3-bit magnitude comparators, the prototype achieves deterministic, high-speed, and verifiable credential evaluation with zero heuristic ambiguity.

Future Scope:

1. HDL Synthesis: Translating the combinational netlist into Verilog/VHDL for deployment on Xilinx Spartan-7 or Intel Cyclone V FPGA chips.
2. Hardware Microcontroller Interface: Interfacing with an ATmega328P / ARM Cortex-M microcontroller with physical 7-segment displays and RGB status LEDs.
3. Expanded State Space: Incorporating entropy variance into an 8-variable ($2^8 = 256$ minterm) extended logic array.

6. References

- [1] M. Morris Mano and Michael D. Ciletti, 'Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog', 6th Edition, Pearson Education, 2018.
- [2] National Institute of Standards and Technology (NIST), 'Digital Identity Guidelines: Authentication and Lifecycle Management', NIST Special Publication 800-63B, 2020.
- [3] John F. Wakerly, 'Digital Design: Principles and Practices', 5th Edition, Pearson, 2017.
- [4] Flask Web Development Framework Documentation, Pallets Projects, 2026. [Online]. Available: <https://flask.palletsprojects.com/>.

Submission Checklist

- [X] PDF report uploaded on the Project Portal (Submit Work page)
- [X] YouTube demo video link added (Unlisted visibility)
- [X] GitHub repository link added, with README.md
- [X] Hardcopy of the report submitted to the department, if required